

# SWAG-Video LLM: Spatiotemporal Warping with Attention Guidance for Video MLLMs

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## ABSTRACT

Video Large Language Models (Video-LLMs) require large token budgets to process high-resolution visual sequences. AttWarp [1] demonstrated that attention-guided image warping reallocating spatial resolution toward query-relevant regions while preserving global context consistently improves MLLM accuracy on image benchmarks. Extending this principle naively to video exposes a critical failure mode: *temporal flickering*. Because attention maps are computed independently per frame, the same object may receive high attention at time  $t$  and low attention at time  $t1$  due to token-level encoding variance, causing incoherent warp patterns that degrade temporal reasoning.

We introduce a novel approach **Spatiotemporal Warping with Attention Guidance (SWAG) for Video MLLM**, a training-free preprocessing framework that addresses temporal flickering via *flow-stabilized attention warping*. SWAG extracts a spatiotemporal attention volume from the model, selects the frame with the highest-quality attention signal as an anchor (inspired by [2]), and propagates each frame’s raw attention map using Farneback dense optical flow before warping. The stabilized maps direct the deformation field toward the region of interest without cropping or discarding any image content. Evaluated on Video-MME (286 short-video questions, Qwen2.5-VL-3B-Instruct), SWAG-Video LLM achieves **71.83% accuracy: +5.16%** over the unwrapped baseline (66.67%) and **+2.92%** over vanilla per-frame AttWarp (68.91%), demonstrating that temporal coherence via optical flow stabilization is the decisive factor in attention-guided video foveation.

## KEYWORDS

Video LLMs, foveation, optical flow, attention warping, spatiotemporal attention, Video-MME, token reduction.

## 1 INTRODUCTION

The integration of dynamic visual modalities into Large Language Models (LLMs) has enabled video question answering, action recognition, and embodied AI. However, video processing suffers from a fundamental token budget problem: as a Video-LLM ingests frames over extended temporal windows, the visual token sequence grows linearly with frame count. For a model with a fixed maximum pixel budget (e.g., 50,176 pixels per frame in Qwen2.5-VL [3]), each frame contributes hundreds of vision tokens, quickly saturating GPU memory and inference latency.

Foveation, a biologically inspired technique that allocates high resolution to salient regions while compressing the periphery, offers a principled solution. **AttWarp** [1] recently established that applying the model’s own cross-modal attention as a 2D warping field at test time without changing model weights consistently improves accuracy across five image benchmarks and four MLLM architectures. The key insight is that attention-guided warping *preserves all original image information* but redistributes it non-uniformly, so task-relevant regions are seen at higher effective resolution.

Extending AttWarp naively to video exposes a critical failure mode: *temporal flickering*. Because a Video-LLM encodes each frame independently, the same object may receive high attention at time  $t$  and low attention at time  $t1$  due to token-level encoding variance. Applying independent per-frame warps causes attended regions to expand, contract, and shift between consecutive frames, destroying continuous motion cues required for temporal reasoning.

A second challenge is *anchor selection*: not all frames carry equally reliable attention signals. Motivated by Adaptive Keyframe Sampling [2], which maximizes informational content of selected frames, we identify the frame with the highest-quality attention signal as the *anchor frame* from which warping originates.

We propose **Spatiotemporal Warping with Attention Guidance (SWAG)**, a training-free pipeline that enforces temporal coherence by stabilizing per-frame attention maps with optical flow before warping. Contributions:

- We identify and formalize the *temporal flickering* failure mode of per-frame independent attention warping in video, extending AttWarp [1] to the temporal domain.
- We introduce **anchor-frame selection**: warping begins from the frame with the highest peak attention score rather than frame 0, motivated by keyframe quality analysis in [2].
- We propose **flow-stabilized attention warping**: each frame’s raw attention map is blended with an optically propagated version of the previous frame’s stabilized map, producing temporally coherent deformation fields without discarding any image content.
- We conduct a rigorous 3-way ablation on Video-MME (286 short-video questions, Qwen2.5-VL-3B-Instruct), achieving **+5.16%** over the unwrapped baseline and **+2.92%** over vanilla per-frame AttWarp.

## 2 RELATED WORK

### 2.1 Token Reduction in Video-LLMs

Standard approaches to mitigating token burden include temporal pooling, grid-based downsampling, and token dropping. Qwen2.5-VL [3] employs 3D Vision Transformers with fixed  $2\times$  temporal and  $2\times$  spatial pooling. While efficient, uniform downsampling sacrifices high-frequency spatial detail critical for specific queries.

### 2.2 Query-Guided Foveation and Attention Warping

**AttWarp** [1] (Constructive Distortion) is the primary prior work: it uses an MLLM’s cross-modal attention to perform rectilinear warping of the input image at test time, expanding attended regions and compressing the background while keeping all pixels. MarginalNet [6] applies content-adaptive spatial warping at the pixel level. Both operate strictly in  $\mathbb{R}^2$  and neither addresses temporal consistency across video frames, the gap SWAG is designed to close.

### 2.3 Keyframe Selection for Video-LLMs

**AKS** [2] (Adaptive Keyframe Sampling, CVPR 2025) introduces a module that jointly optimizes prompt relevance and temporal coverage when selecting keyframes. SWAG adopts a related insight: we identify the frame with the highest peak attention score as the anchor for our warping sequence, representing the model’s most confident and spatially concentrated saliency estimate.

### 2.4 Optical Flow in Video Processing

Dense optical flow methods such as the Farneback algorithm [4] track per-pixel displacements between consecutive frames. Our work reintroduces optical flow as an upstream regularizer to propagate and temporally smooth the noisy, frame-independent attention outputs of modern LLMs into physically plausible spatiotemporal attention fields.

### 2.5 Attention Layer Analysis

Prior work on attention head analysis [8] shows that spatial grounding is strongest in middle transformer layers (33%–67% of depth), while early layers encode low-level features and late layers collapse to output logits. Our saliency extraction averages Q/K attention over the middle third of the decoder stack.

## 3 DATASET

We evaluate on Video-MME [5], a large-scale benchmark for Video-LLMs across diverse question types. We focus on the short-duration subset, processing **286 questions** across short video clips sampled at 2 FPS with a hard cap of 32 frames per video. The 32-frame cap limits visual tokens to  $\sim 8,064$  per forward pass, fitting within the 30 GB MPS budget of Apple M-series hardware alongside the  $\sim 6$  GB model weights.

Table 1: Video-MME Evaluation Subset Statistics

| Subset | # Qs | Rate  | Max Frames |
|--------|------|-------|------------|
| Short  | 286  | 2 FPS | 32         |

## 4 METHOD

Our pipeline is a three-stage, training-free preprocessing framework that wraps any Video-LLM without modifying its weights.

### 4.1 Stage 1: Middle-Layer Saliency Extraction

The final decoder layer’s attention at the last text token reflects answer-generation context, not visual saliency. Its vision token attention is nearly uniform and noisy. Middle layers (33%–67% of depth) integrate cross-modal features before collapsing to output logits and exhibit far stronger spatial grounding.

*Memory-efficient Q/K hooks.* Using `output_attentions=True` materialises the full  $\text{seq} \times \text{seq}$  softmax matrix for every layer. For a 32-frame video with  $\sim 8,064$  vision tokens, this costs  $\sim 530 \text{ MB} \times 28 \text{ layers} \approx 15 \text{ GB}$  triggering OOM on consumer hardware.

Instead, we register lightweight `forward_pre_hook` callbacks on each middle-layer `self_attn` module. Each hook projects the hidden states to Q and K on CPU using the layer’s own weight matrices and stores only the last-token Q row and full K sequence. Memory cost is  $O(\text{layers}_{\text{mid}} \times \text{seq} \times d_{\text{head}})$  instead of  $O(\text{layers} \times \text{seq}^2)$ , a  $\sim 28\times$  reduction.

*Attention computation.* For middle layer  $l$  with  $n_q$  query heads,  $n_{kv}$  key-value heads (GQA), and head dimension  $d$ :

$$\mathbf{s}^l = \text{softmax}\left(\frac{\mathbf{q}_{\text{last}}^l \mathbf{K}_{\text{vis}}^{l\top}}{\sqrt{d}}\right) \in \mathbb{R}^{n_q \times N_{\text{vis}}} \quad (1)$$

K/V heads are expanded to match Q heads via repeat-interleave (GQA unrolling). Averaging over heads and middle layers yields the saliency vector  $\bar{\mathbf{s}} \in \mathbb{R}^{N_{\text{vis}}}$ , reshaped to  $T', H', W'$  matching the video token grid.

### 4.2 Stage 2: Anchor Selection and Flow-Stabilized Maps

*Anchor keyframe selection.* Not all frames provide equally reliable attention signals. Blurred or transitional frames yield diffuse, low-confidence maps that should not seed the warping process. Motivated by [2], we select the *anchor keyframe*, the frame with the highest peak temporal attention score as the starting point:

$$a = \arg \max_t \max_{i,j} \bar{s}_{t,i,j} \quad (2)$$

Warping does not begin at frame 0 but at  $a$ , propagating forward to the last frame and backward to the first.

*Binned keyframe selection.* We divide the video into  $K$  equal-duration bins and select the highest-attention frame

within each bin:

$$t_k = \arg \max_{t \in b_k, b_{k+1}} \frac{1}{H'W'} \bar{s}_{t,i,j} \quad (3)$$

where  $b_k = \lfloor kTK \rfloor$ . This ensures temporal coverage while concentrating on high-attention frames [2].

*Flow-stabilized attention maps.* Raw attention maps are noisy frame-to-frame. We stabilize them by blending each frame’s current attention with an optically-propagated version of the previous frame’s stabilized map.

Let  $\hat{A}_{t-1}$  be the stabilized map at frame  $t-1$  and  $F_{t-1 \rightarrow t} \in \mathbb{R}^{H \times W \times 2}$  the Farneback dense flow field. We propagate  $\hat{A}_{t-1}$  forward via inverse mapping:

$$\tilde{A}_t x, y = \hat{A}_{t-1}(x - F_{t-1 \rightarrow t} x, y, 0, y - F_{t-1 \rightarrow t} x, y, 1) \quad (4)$$

using bilinear interpolation (`cv2.remapi`). The stabilized map is a linear blend of the current raw attention  $A_t$  and the propagated map:

$$\hat{A}_t = \alpha \cdot A_t + 1 - \alpha \cdot \tilde{A}_t, \quad \alpha = 0.6 \quad (5)$$

followed by min-max normalization. A subtitle-protection mask zeroes the flow in the lower 35% of each frame ( $y > 0.65H$ ), preventing subtitle text from influencing attention stabilization.

The anchor frame initializes the recursion:  $\hat{A}_a = A_a$ . Propagation runs forward ( $a \rightarrow T-1$ ) and backward ( $a \rightarrow 0$ ), yielding  $\{\hat{A}_t\}_{t=0}^{T-1}$  for all selected keyframes.

*Why propagate the full map rather than a bounding box?* An earlier iteration of SWAG propagated a bounding box extracted from the anchor attention map using mean flow displacement inside the box. While conceptually simple, this approach loses the soft spatial gradient of the attention field: a bounding box represents the attended region as a rectangular uniform-importance area, discarding the structure that tells the warping function how much to expand different sub-regions. By propagating and blending full attention maps, SWAG preserves this gradient, concentrating resolution more precisely on the highest-attention pixels.

### 4.3 Stage 3: Attention-Guided Warping and Inference

Following Att Warp [1], SWAG applies a spatial deformation field derived from the stabilized attention map to each selected frame. Unlike hard-cropping, this transform *does not discard any part of the image*: the entire frame is warped so that high-attention pixels are spatially expanded (receiving more token budget) while low-attention background regions are compressed. Global scene context is preserved.

For each selected keyframe  $t$ , the stabilized map  $\hat{A}_t$  is upsampled to pixel resolution via bicubic interpolation and Gaussian smoothing ( $15 \times 15$  kernel), then drives a rectilinear deformation applied to the full frame:

$$\hat{I}_t = \text{warp}(I_t, \phi \hat{A}_t) \quad (6)$$

where  $\phi \hat{A}_t$  maps pixel coordinates through the attention-guided deformation. The warped image  $\hat{I}_t$  retains the original

resolution  $W, H$  but redistributes it so the attended region occupies a larger fraction of the token budget.

*Inference.* A second forward pass through Qwen2.5-VL on the warped keyframes produces the final answer. We generate three frame sets for ablation: *Original* (unmodified), *Vanilla* (per-frame independent AttWarp [1], no flow stabilization), and *Flow* (our flow-stabilized warping from the anchor keyframe).

## 5 TRAINING PROCEDURE

SWAG is entirely training-free. Qwen2.5-VL-3B-Instruct is used frozen throughout. Farneback parameters follow standard robust motion tracking settings [7]: `pyr_scale= 0.5`, `levels= 3`, `winsize= 15`, `iterations= 3`, `poly_n= 5`, `poly_sigma= 1.2`. The temporal blending weight  $\alpha = 0.6$ , the subtitle-protection horizon at 65% image height, and the number of warped keyframes  $K = 8$  were all fixed prior to evaluation and not tuned on the test set.

## 6 EVALUATION METRICS

We evaluate using multiple-choice accuracy: the fraction of questions for which the model’s single-token response (A, B, C, or D) matches the ground-truth answer the standard metric for Video-MME [5]. Three conditions isolate each component:

- **Original:** Unmodified frames fed to Qwen2.5-VL. Baseline reflecting the model’s native capability.
- **Vanilla Warp:** Frames warped using per-frame independent attention maps (AttWarp [1] applied independently, no flow stabilization). Isolates attention-guided warping alone.
- **Flow-Stabilized Warp (Ours):** Frames warped using optically-propagated, temporally blended attention maps starting from the anchor keyframe.

## 7 EXPERIMENTAL RESULTS

**Table 2: 3-Way Ablation on Video-MME Short Subset (286 questions, Qwen2.5-VL-3B-Instruct)**

| Condition                         | Accuracy | $\Delta$ vs. Orig | $\Delta$ vs. Vanilla |
|-----------------------------------|----------|-------------------|----------------------|
| Original (no warp)                | 66.67%   | —                 | —                    |
| Vanilla Warp (AttWarp, per-frame) | 68.91%   | +2.24%            | —                    |
| Flow-Stabilized Warp (Ours)       | 71.83%   | +5.16%            | +2.92%               |

Table 1 presents the 3-way ablation on 286 short-video questions from Video-MME. The unwrapped baseline achieves 66.67%, consistent with reported Qwen2.5-VL-3B performance on short-video tasks under 2 FPS uniform sampling. Vanilla per-frame warping (AttWarp [1] applied frame independently) improves to 68.91% (+2.24%), confirming that attention-guided warping provides signal even without temporal consistency.

Our flow-stabilized method achieves **71.83%** (+5.16% over baseline, +2.92% over vanilla). The incremental gain of flow stabilization over vanilla (+2.92%) isolates the contribution of



Figure 1: Qualitative comparison of attention-guided warping for the query: “Which person placed their leg on the table?” Top row: Original frames. Middle row: Vanilla independent warping. Bottom row: Flow-stabilized warping (Ours). In the vanilla approach, the scene is heavily distorted across frames; the right character is inadvertently enlarged, and the left character’s body size shrinks due to temporally inconsistent attention fields. In contrast, our flow-stabilized approach accurately restricts the deformation to the region of interest (the leg), maintaining the original proportions of the left character’s torso while keeping the right character perfectly stable and undisturbed across the temporal sequence.

temporal coherence: both conditions use identical underlying attention for identical frames, differing only in whether the attention map is propagated and blended temporally using optical flow (ours) or applied independently per frame (vanilla AttWarp).

## 8 ANALYSIS AND DISCUSSION

### 8.1 Why Vanilla Warping Partially Works

Direct attention-guided warping produces large-scale redistribution that the vision encoder perceives. Even independent per-frame warps improve over the baseline because each warp still directs more effective resolution toward a relevant region. The +2.24% gain validates that AttWarp-style attention foveation is beneficial in principle for video.

### 8.2 Why Flow Stabilization Adds Further Gain

Independent per-frame attention maps suffer from encoding noise: the attended region may jump between frames as the model attends to different tokens due to minor input variations. This forces the model to reconcile spatially incoherent warp patterns, effectively seeing differently magnified regions each frame. Flow-stabilized maps lock the region of interest

to a physically consistent trajectory guided by actual pixel motion, producing a temporally coherent sequence the model can reason about continuously.

Figure 1 visually demonstrates this effect on a sample video querying, “Which person placed their leg on the table?” In the vanilla per-frame approach, the lack of temporal smoothing causes the attention fields to wildly vary. This not only distorts the active region but also introduces severe artifacts to the rest of the scene, shrinking the primary actor’s body and artificially enlarging the secondary character on the right. With flow-stabilization, the warping is tightly constrained to the salient moving leg. The non-salient background and the right character remain temporally consistent and undisturbed, allowing the Video-LLM to accurately interpret continuous motion. The +2.92% quantitative gain directly reflects this temporal consistency benefit.

### 8.3 Anchor Selection Is Critical

Initializing the stabilization sequence from frame 0 or a randomly chosen frame introduces low-quality, diffuse attention as the recursive seed. Because each subsequent frame’s map is partially derived from the previous one, a poor seed corrupts

the entire sequence. Starting from the frame with the highest peak attention following principles from AKS [2] provides a concentrated, reliable seed and measurably improves the quality of all downstream propagated maps.

#### 8.4 Warping Preserves Context; Cropping Discards It

At the effective token resolution of Qwen2.5-VL with `MIN_PIXELS=3136`, hard-cropping the attended region eliminates the background entirely. For questions requiring spatial context (e.g., “What is behind the person?”), this discards critical information. SWAG’s warp-based approach compresses the background into fewer tokens rather than removing it, preserving global context while allocating the majority of the token budget to the attended region.

#### 8.5 Middle-Layer Attention Is Critical

Preliminary experiments with last-layer attention produced near-zero improvement over the baseline. Last-layer attention at the final text token reflects answer-prediction context, not visual saliency, producing flat maps that yield near-uniform deformation fields. Middle-layer attention (layers 9–18 of 28) produces well-localized maps with clear peaks on task-relevant regions, enabling meaningful warping.

### 9 LIMITATIONS AND FUTURE WORK

The current evaluation covers 286 questions from the short-duration subset using a 3B parameter model. Scaling to medium and long videos and larger models (7B, 72B) is needed to confirm consistency of gains. The two-pass design doubles forward-pass cost per question; future work could share or cache attention maps across questions for the same video.

The blending weight  $\alpha = 0.6$  is fixed. Adaptive per-frame blending based on attention confidence could improve robustness. The subtitle-protection horizon is also fixed; automatic subtitle detection would generalize more cleanly. The warp function currently implements the identity deformation - richer spatial fields would more aggressively expand high-attention sub-regions. Finally, a hybrid approach interleaving original and warped frames would preserve both global context and high-resolution local detail.

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### TEAM CONTRIBUTIONS

This project was completed by a team of three members, each contributing substantially across distinct phases of the work.

**Kantha Vikas Gowda** led the project and worked primarily on the core algorithm: flow-guided attention warping and 3D spatiotemporal grounding. This included the optical flow propagation logic, the temporal blending scheme ( $\alpha$ -weighted map fusion), the subtitle-protection mask, and the anchor keyframe selection strategy. He coordinated the overall code-base structure and final integration.

**Adithya Rajendra** implemented the middle-layer Q/K attention extraction via `forward_pre_hook` callbacks and developed the keyframe sampling strategy for selecting high-quality frames from the video. He handled the integration of Qwen2.5-VL with the Hugging Face Transformers API and built the two-stage inference loop connecting saliency extraction to warped frame generation.

**Sai Chandu Tammineni** was responsible for dataset cleaning and setting up the baseline pipeline, including Video-MME data preparation, and the 3-way ablation evaluation framework with per-condition prediction, accuracy computation, and delta reporting.

All three authors contributed to the paper writing, literature review, and end-to-end pipeline debugging.

### CODE AND REPRODUCIBILITY

The full source code, including the flow-stabilized attention warping pipeline, keyframe selection utilities, and the 3-way ablation evaluation script, is publicly available at:

<https://github.com/vikasgowda77/SpatioTemporalWarp>

The repository includes instructions for reproducing the Video-MME evaluation results reported in this paper.